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HYBRID PIXEL AND HYBRID SENSOR

Abstract

Provided is a hybrid pixel including a semiconductor substrate, a pixel circuit including at least one transfer transistor, a sensing circuit configured to detect movement of an object, and at least one photodiode having one end connected to the pixel circuit and another end connected to the sensing circuit, the semiconductor substrate having a first conductivity type, the semiconductor substrate including a photoelectric conversion area corresponding to the at least one photodiode, the photoelectric conversion area including a first area of a second conductivity type and a second area, the second area is physically separated from a second area of an adjacent hybrid pixel by a deep trench isolation area, and the semiconductor substrate including the deep trench isolation area.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application Nos. 10-2024-0032844, filed on Mar. 7, 2024, and 10-2024-0083742, filed on Jun. 26, 2024, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entirety.

BACKGROUND

[0002] The inventive concepts relate to hybrid pixels and hybrid sensors. More particularly, the inventive concepts relate to hybrid pixels and hybrid sensors, which are capable of performing both a function of an image sensor and a function of a dynamic vision sensor.

[0003] An image sensor is a semiconductor device that converts an optical signal incident from the outside into an electrical signal, and generates image information corresponding to the incident optical signal. With the recent development of computer industries and communication industries, demands for image sensors have increased in various fields such as digital cameras, camcorders, a mobile phone, surveillance cameras, and medical micro-cameras. An image sensor may be capable of generating an image of excellent image quality however, the image sensor may also consume a lot of power.

[0004] A vision sensor, for example, a dynamic vision sensor, generates information about an event, e.g., an event signal, when the event (e.g., a change in intensity of light) occurs, and transmits the event signal to a processor. A vision sensor has low power consumption because a result is output only in a pixel where a signal change has occurred, but is able to output only information about an event.

[0005] Recently, there have been attempts to achieve a fast frame rate with high image quality by combining an image of an image sensor and an image of a dynamic vision sensor. When a portion of a pixel area is used as a dynamic vision sensor, signal-to-noise ratio (SNR) deterioration may occur due to a decrease in the amount of light and it may be difficult to design pixels.

SUMMARY

[0006] The inventive concepts provide hybrid sensors capable of using both holes and electrons of a photodiode.

[0007] According to some aspects of the inventive concepts, there is provided a hybrid pixel.

[0008] The hybrid pixel includes a semiconductor substrate, a pixel circuit including at least one transfer transistor, a sensing circuit configured to detect movement of an object, and at least one photodiode having one end connected to the pixel circuit and another end connected to the sensing circuit, the semiconductor substrate having a first conductivity type, the semiconductor substrate including a photoelectric conversion area corresponding to the at least one photodiode, the photoelectric conversion area including a first area of a second conductivity type and a second area surrounding the first area, the second area is physically separated from a second area of an adjacent hybrid pixel by a deep trench isolation area, and the semiconductor substrate including the deep trench isolation area.

[0009] According to some aspects of the inventive concepts, there is provided a hybrid pixel.

[0010] The hybrid pixel includes a semiconductor substrate, a pixel circuit including at least one transfer transistor, a sensing circuit configured to detect movement of an object, and at least one

photodiode having one end connected to the pixel circuit and another end connected to the sensing circuit, the semiconductor substrate having a first conductivity type, the semiconductor substrate including a photoelectric conversion area corresponding to the at least one photodiode, the photoelectric conversion area including a first area of a second conductivity type and a second area, the second area physically separated from a second area of an adjacent hybrid pixel by a deep trench isolation area, and a first layer including the pixel circuit and the at least one photodiode is different from a second layer including the sensing circuit.
[0011] According to some aspects of the inventive concepts, there is provided a hybrid sensor.
[0012] The hybrid sensor includes a pixel array including a plurality of hybrid pixels, the plurality of hybrid pixels including a photodiode, a pixel circuit connected to one end of the photodiode, a sensing circuit connected to another end of the photodiode, and a noise removal circuit connected between the sensing circuit and the other end of the photodiode, and each of the plurality of hybrid pixels is physically separated from an adjacent hybrid pixel by a deep trench isolation area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0013] Example embodiments will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which:
[0014] FIG. **1** is a block diagram of an image processing apparatus according to some example embodiments;
[0015] FIG. **2** is a block diagram of a hybrid sensor according to some example embodiments;
[0016] FIGS. **3**A, **3**B, and **3**C are diagrams illustrating implementation examples of a pixel array corresponding to a color filter array, according to some example embodiments;
[0017] FIG. **4**A is a diagram for describing a circuit diagram of a hybrid pixel, according to some example embodiments;
[0018] FIG. **4**B illustrates some example embodiments of a circuit diagram of the hybrid pixel of FIG. **4**A;
[0019] FIG. **5**A is a cross-sectional view of a layout region corresponding to an A region of FIG. **4**B;
[0020] FIG. **5**B is a diagram for describing a flow of charges in the cross-sectional view of FIG. **5**A;
[0021] FIG. **6**A is a diagram for describing a circuit diagram of a hybrid pixel, according to some example embodiments;
[0022] FIG. **6**B is a layout plan view illustrating some components of the hybrid pixel of FIG. **6**A;
[0023] FIG. **6**C is a cross-sectional view taken along a line A-A′ of FIG. **6**B;
[0024] FIG. **7** is a circuit diagram of a hybrid pixel according to some example embodiments;
[0025] FIG. **8** illustrates a stack structure of a hybrid sensor, according to some example embodiments;
[0026] FIG. **9** illustrates a stack structure of a hybrid sensor, according to some example embodiments;
[0027] FIG. **10**A illustrates a stack structure of components of a hybrid pixel included in a hybrid sensor, according to some example embodiments;
[0028] FIG. **10**B illustrates a stack structure of components of a hybrid pixel included in a hybrid sensor, according to some example embodiments;
[0029] FIG. **11** is a block diagram of a structure of a hybrid pixel, according to some example embodiments;
[0030] FIG. **12** is a circuit diagram of a hybrid pixel, according to some example embodiments;
[0031] FIG. **13** is a timing diagram for describing turn-on timing of transistors included in a noise

removal circuit of FIG. **12**;

[0032] FIG. **14** is a timing diagram for describing turn-on timing of transistors included in a noise removal circuit according to some example embodiments; and

[0033] FIG. **15** is a timing diagram for describing turn-on timing of transistors included in a noise removal circuit according to some example embodiments.

DETAILED DESCRIPTION

[0034] Hereinafter, various embodiments will be described with reference to accompanying drawings.

[0035] FIG. **1** is a block diagram of an image processing apparatus **10** according to some example embodiments.

[0036] Referring to FIG. **1**, the image processing apparatus **10** may include a hybrid sensor **100** and a processor **300**. The image processing apparatus **10** according to some example embodiments may be mounted on an electronic device having an image or optical sensing function. For example, the image processing apparatus **10** may be mounted on an electronic device such as a camera, a smartphone, a wearable device, an Internet of things (IoT) device, a tablet personal computer (PC), a personal digital assistant (PDA), a portable multimedia player (PMP), a navigation device, a drone, or an advanced driver assistance system (ADAS). Also, the image processing apparatus **10** may be included, as a component, in a vehicle, furniture, manufacturing equipment, a door, or any type of measuring device.

[0037] The hybrid sensor **100** may be a sensor capable of performing both a vision sensor function and an image sensor function. The vision sensor function may be a function of outputting an event signal by detecting a change in intensity of incident light. The vision sensor function provided by the hybrid sensor **100** may be a dynamic vision sensor function of outputting event signals for pixels where a change in intensity of light has been detected, e.g., for pixels where an event has occurred. The change in intensity of light may be caused by movement of an object photographed by the hybrid sensor **100** or caused by movement of the hybrid sensor **100** or image processing apparatus **10** itself. The hybrid sensor **100** may periodically or aperiodically transmit, to the processor **300**, pieces of vision sensor data VDT including the event signals.

[0038] The image sensor function is a function of converting an optical signal of an object, which is incident through an optical lens, into an electrical signal, generating image data IDT based on the electrical signals, and outputting the image data IDT. The hybrid sensor **100** may include, for example, a readout circuit and a pixel array including a plurality of pixels arranged two dimensionally, and the pixel array may convert received optical signals into electrical signals. The pixel array may be implemented with a photoelectric conversion element, for example, a charge coupled device (CCD) or a complementary metal oxide semiconductor (CMOS), or may be implemented with any type of photoelectric conversion element. The readout circuit may generate raw data based on the electrical signal provided from the pixel array and output, as the image data IDT, the raw data or raw data on which preprocessing such as bad pixel removal has been performed. The hybrid sensor **100** may be implemented with a semiconductor chip or package including the pixel array and the readout circuit.

[0039] According to the inventive concepts, the hybrid sensor **100** may perform both the image sensor function and the vision sensor function respectively based on pieces of information (e.g., electrons and holes) obtained from at least one photodiode included in a hybrid pixel. The hybrid pixel included in the pixel array included in the hybrid sensor **100** may include the at least one photodiode, a pixel circuit connected to one end of the photodiode, and a sensing circuit connected to another end of the photodiode.

[0040] The processor **300** may perform image processing on the image data IDT provided from the hybrid sensor **100**. For example, the processor **300** may perform image processing of changing a data format of the image data IDT (e.g., changing image data of a Bayer pattern to a YUV or RGB format), image processing for image quality improvement, such as noise removal, brightness adjustment, or sharpness adjustment, and/or the like. The processor **300** may process the vision sensor data VDT received from the hybrid sensor **100** and detect movement of the object (or movement of the object in an image recognized by the image processing apparatus **10**), based on the event signal in the vision sensor data VDT.

[0041] Also, the processor **300** may match an image frame included in the image data IDT provided from the hybrid sensor **100** with the vision sensor data VDT received from the hybrid sensor **100**, based on a timestamp and pieces of synchronization signal information. The processor **300** may include an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a dedicated microprocessor, a microprocessor, or a general purpose processor. According to some example embodiments, the processor **300** may be an application processor or an image signal processor.

[0042] The hybrid sensor **100** and the processor **300** may each be implemented as an integrated circuit (IC). For example, the hybrid sensor **100** and the processor **300** may be implemented as separate semiconductor chips. Alternatively, the hybrid sensor **100** and the processor **300** may be implemented as a single chip. For example, the hybrid sensor **100** and the processor **300** may be implemented as a system-on-chip (SoC).

[0043] The image processing apparatus **10** may control an external device **400** and collect data. The device **400** may include an acceleration sensor, an inertial measurement unit (IMU), a gyro sensor, an infrared ray (IR) light-emitting diode (LED), and a flash light.

[0044] The acceleration sensor is a sensor configured to measure an acceleration of a moving object or strength of an impact, and may measure dynamic force, such as an acceleration, vibration, or impact, of an object by processing an output signal. The gyro sensor is a sensor used for location measurement and direction setting by using dynamic motion of a rotating object. The IR LED is for capturing an image when there is no light, and is a device used in a closed-circuit television (CCTV) and/or the like.

[0045] The IMU uses a combination of an accelerometer, a gyroscope, and sometimes a magnetometer, and is recently operates as an orientation sensor in many consumer products, such as a mobile phone and a camera. The IMU operates by detecting linear acceleration by using one or more accelerometers and detecting a rotating speed by using one or more gyroscopes, and may include a magnetometer in some cases. In a general configuration, the IMU may include one accelerometer, one gyroscope, and one magnetometer per axis for three axes of pitch, roll, and yaw.

[0046] FIG. **2** is a block diagram of the hybrid sensor **100** according to some example embodiments.

[0047] The hybrid sensor **100** may include a pixel array **110**, a row driver **120**, a readout circuit **130**, a ramp signal generator **140**, a timing controller **150**, an event detection circuit **160**, and an interface circuit **170**, and the readout circuit **130** may include an analog-to-digital conversion (ADC) circuit **131** and a data bus **132**.

[0048] The pixel array **110** includes a plurality of row lines RL, a plurality of column lines CL, and a plurality of pixels PX accessed to the plurality of row lines RL and the plurality of column lines CL, and arranged in a matrix. The plurality of pixels PX may be hybrid pixels. According to some example embodiments, each of the plurality of pixels PX may include a pixel circuit configured to output an image signal and a sensing circuit configured to sense whether an event has occurred. According to some example embodiments, the sensing circuit may detect movement of an object by detecting the movement as an event. The pixel circuit and the sensing circuit included in each of the plurality of pixels PX may share a photodiode included in each of the plurality of pixels PX. According to some example embodiments, the pixel circuit may operate based on electrons of the photodiode and the sensing circuit may operate based on holes of the photodiode. A structure corresponding to the plurality of pixels PX will be described below with reference to FIG. **4**A.

[0049] Each pixel PX may include at least one photoelectric conversion element. The pixel PX may detect light by using the photoelectric conversion element, and output an image signal that is an electrical signal according to the detected light. For example, the photoelectric conversion element may be a light detection element including an organic material or an inorganic material, such as an inorganic photodiode, an organic photodiode, a perovskite photodiode, a phototransistor, a photogate, and/or a pinned photodiode. According to some example embodiments, each pixel PX may include a plurality of photoelectric conversion elements.

[0050] A micro-lens (not shown) for focusing light may be arranged above each pixel PX or above each of pixel groups including adjacent pixels PX. Each of the plurality of pixels PX may detect light of a specific spectrum domain from light received through the micro-lens arranged thereabove. For example, the pixel array **110** may include a red I pixel converting light of a red spectrum domain into an electrical signal, a green (G) pixel converting light of a green spectrum domain into an electrical signal, a blue (B) pixel converting light of a blue spectrum domain into an electrical signal, and a white (W) pixel for noise adjustment. A color filter array CF for transmitting light of a specific spectrum domain may be arranged above each of the plurality of pixels PX. This will be described below with reference to FIGS. **3**A to **3**C. However, some example embodiments is not limited thereto and the pixel array **110** may include pixels converting light of a spectrum domain other than red, green, and blue, into an electrical signal. According to some example embodiments, the pixel PX included in the pixel array **110** may be an RGBW pixel.

[0051] Charges generated by the photoelectric conversion element, such as the photodiode, in each pixel PX may be accumulated in a floating diffusion node, and the charges accumulated in the floating diffusion node may be converted into a voltage. A ratio at which the charges accumulated in the floating diffusion node are converted into a voltage may be referred to as a conversion gain. The conversion gain may vary according to capacitance of the floating diffusion node.

[0052] The row driver **120** may drive the pixel array **110** in units of row lines RL. The row driver **120** may select at least one row line RL from among the row lines RL configuring the pixel array **110**. For example, the row driver **120** may generate a selection signal of selecting one of the plurality of row lines RL. The pixel array **110** may output a pixel signal from the row line RL selected according to the selection signal. The pixel signal may include a reset signal and an image signal.

[0053] The row driver **120** may generate control signals for controlling the pixel array **110**. For example, the row driver **120** may generate control signals for controlling transistors included in the sensing circuit and pixel circuit included in the pixel PX. The row driver **120** may independently (for example, alone or without direction or interference from another element/feature) provide the control signals for controlling the transistors included in the sensing circuit and pixel circuit included in the pixel PX. The row driver **120** may provide the control signals to the plurality of pixels PX, in response to a timing control signal provided by the timing controller **150**.

[0054] The timing controller **150** may control timings of the row driver **120**, readout circuit **130**, and ramp signal generator **140**. The timing controller **150** may provide control signals for controlling operating timings of the row driver **120**, the readout circuit **130**, and the ramp signal generator **140**, respectively. The timing controller **150** may adjust timings of a plurality of control line signals generated by the row driver **120** to determine activating and deactivating timings of signals applied to a control line. A detailed timing control method of the timing controller **150** will be described below.

[0055] The ramp signal generator **140** may generate a ramp signal RAMP that is increased or decreased in a certain inclination, and provide the ramp signal RAMP to the ADC circuit **131** of the readout circuit **130**.

[0056] The readout circuit **130** may read out a pixel signal from the pixels PX of the row line RL selected by the row driver **120**, from among the plurality of pixels PX. The readout circuit **130** may convert pixel signals received from the pixel array **110** through the plurality of column lines CL into digital data, based on the ramp signal RAMP provided from the ramp signal generator **140**, and generate and output pixel values corresponding to the plurality of pixels PX in units of rows.

[0057] The ADC circuit **131** may compare each of the pixel signals received through column lines CL with the ramp signal RAMP and generate a pixel value that is a digital signal, based on results of the comparison. For example, the reset signal may be removed from the image signal and a pixel value indicating the amount of light detected in the pixel PX may be generated. The ADC circuit **131** may sample and hold the pixel signal according to a correlated double sampling (CDS) method and double-sample a level of specific noise (e.g., the reset signal) and a level according to the image signal, thereby generating a comparison signal based on a level corresponding to a difference therebetween. The ADC circuit **131** may first read out the image signal and sample the pixel signal provided by reading out the reset signal, according to a delta reset sampling (DRS) method.
[0058] The plurality of pixel values generated by the ADC circuit **131** may be output through the data bus **132**, as the image data IDT. The image data IDT may be provided to an image signal processor inside or outside the hybrid sensor **100**.
[0059] The data bus **132** may temporarily store the pixel value output from the ADC circuit **131** and then output the same. The data bus **132** may include a plurality of column memories and a column decoder. The plurality of pixel values stored in the plurality of column memories may be output as the image data IDT, under control by the column decoder.
[0060] According to some example embodiments, each of the plurality of pixels PX may detect events in which intensity of received light is increased or decreased. For example, each of the plurality of pixels PX may be connected to the event detection circuit **160** through a column line extending in a column direction and a row line extending in a row direction. A signal indicating that an event has occurred and polarity information of the event (e.g. whether the event is an on-event in which intensity of light is increased or an off-event in which intensity of light is decreased) may be output from the pixel PX where the event has occurred to the event detection circuit **160**. In the inventive concepts, a signal output from the pixel PX to the event detection circuit **160** is referred to as an event detection signal.
[0061] The event detection circuit **160** may readout events from the pixel array **110** and process the events. The event detection circuit **160** may generate event data EDT including polarity information of an occurred event, an address of a pixel where the event has occurred, and a timestamp. The event detection circuit **160** may process events occurred in the pixel array **110** in units of pixels, in units of pixel groups including a plurality of pixels, in units of columns, or in units of frames.
[0062] The interface circuit **170** may receive the event data EDT and the timestamp, and transmit the vision sensor data VDT to the processor **300** according to a set protocol. The interface circuit **170** may pack the event data EDT and the timestamp in units of individual signals, in units of packets, or in units of frames, according to the set protocol, to generate the vision sensor data VDT, and transmit the vision sensor data VDT to the processor **300** (see FIG. **1**). For example, the interface circuit **170** may include at least one of an address event representation (AER) interface, a mobile industry processor interface (MIPI), and/or a parallel interface.
[0063] According to the inventive concepts, the image signal and the event detection signal may be simultaneously (e.g., at the same or about the same time) output from the pixel PX included in the pixel array **110**. In other words, according to the inventive concepts, the pixel PX included in the pixel array **110** may be a hybrid pixel capable of outputting both the image signal and the event detection signal. Accordingly, desired pieces of image data may be combined and data usability may be increased.
[0064] FIGS. **3**A, **3**B, and **3**C are diagrams illustrating implementation examples of a pixel array corresponding to a color filter array, according to some example embodiments.
[0065] Referring to FIG. **3**A, a pixel array **110***a* includes a plurality of pixels arranged in a plurality of rows and a plurality of columns, and for example, a shared pixel defined in a unit including pixels arranged in two rows and two columns may include four sub-pixels. The pixel array **110***a*

may include first to sixteenth shared pixels SP**0** to SP**15**. The pixel array **110***a* may further include the color filter array CF for the first to sixteenth shared pixels SP**0** to SP**15** to sense various colors. For example, the color filter array CF includes filters sensing rI(R), green (G), and blue (B), and each of the first to sixteenth shared pixels SP**0** to SP**15** may include sub-pixels where a same color filter is arranged. For example, the first shared pixel SP**0**, the third shared pixel SP**2**, the ninth shared pixel SP**8**, and the eleventh shared pixel SP**10** may include sub-pixels including a blue color filter, the second shared pixel SP**1**, the fourth shared pixel SP**3**, the fifth shared pixel SP**4**, the seventh shared pixel SP**6**, the tenth shared pixel SP**9**, the twelfth shared pixel SP**11**, the thirteenth shared pixel SP**12**, and the fifteenth shared pixel SP**14** may include sub-pixels including a green color filter, and the sixth shared pixel SP**5**, the eighth shared pixel SP**7**, the fourteenth shared pixel SP**13**, and the sixteenth shared pixel SP**15** may include sub-pixels including a red color filter. Also, a group including the first shared pixel SP**0**, the second shared pixel SP**1**, the fifth shared pixel SP**4**, and the sixth shared pixel SP**5**, a group including the third shared pixel SP**2**, the fourth shared pixel SP**3**, the seventh shared pixel SP**6**, and the eighth shared pixel SP**7**, a group including the ninth shared pixel SP**8**, the tenth shared pixel SP**9**, the thirteenth shared pixel SP**12**, and the fourteenth shared pixel SP**13**, and a group including the eleventh shared pixel SP**10**, the twelfth shared pixel SP**11**, the fifteenth shared pixel SP**14**, and the sixteenth shared pixel SP**15** may each be arranged in the pixel array **110***a* to correspond to a Bayer pattern. According to some example embodiments, the group including the first shared pixel SP**0**, the second shared pixel SP**1**, the fifth shared pixel SP**4**, and the sixth shared pixel SP**5**, the group including the third shared pixel SP**2**, the fourth shared pixel SP**3**, the seventh shared pixel SP**6**, and the eighth shared pixel SP**7**, the group including the ninth shared pixel SP**8**, the tenth shared pixel SP**9**, the thirteenth shared pixel SP**12**, and the fourteenth shared pixel SP**13**, and the group including the eleventh shared pixel SP**10**, the twelfth shared pixel SP**11**, the fifteenth shared pixel SP**14**, and the sixteenth shared pixel SP**15** may each correspond to a block of the color filter array CF.

[0066] However, this is only an example and the pixel array **110***a* according to some example embodiments may include various types of color filters. For example, the color filter array CF may include filters sensing not only red, green, and blue, but also yellow, cyan, magenta, and white. Also, the pixel array **110***a* may include more shared pixels, and arrangements of the first to sixteenth shared pixels SP**0** to SP**15** may vary.

[0067] Referring to a pixel array **110***b* of FIG. **3**B, each of the first, second, fifth, and sixth shared pixels SP**0**, SP**1**, SP**4**, and SP**5** may include nine sub-pixels. The first shared pixel SP**0** may include nine sub-pixels including a blue (B) color filter, and the second shared pixel SP**1** and the fifth shared pixel SP**4** may each include nine sub-pixels including a green (G) color filter. The sixth shared pixel SP**5** may include nine sub-pixels including led (R) color filter. According to some embodiments, the first, second, fifth, and sixth shared pixels SP**0**, SP**1**, SP**4**, and SP**5** may be referred to as nona cells.

[0068] Referring to a pixel array **110***c* of FIG. **3**C, each of the first, second, fifth, and sixth shared pixels SP**0**, SP**1**, SP**4**, and SP**5** may include sixteen sub-pixels. The first shared pixel SP**0** may include sixteen sub-pixels including a blue (B) color filter, and the second shared pixel SP**1** and the fifth shared pixel SP**4** may each include sixteen sub-pixels including a green (G) color filter. The sixth shared pixel SP**5** may include sixteen sub-pixels includiIa red (R) color filter. According to some embodiments, the first, second, fifth, and sixth shared pixels SP**0**, SP**1**, SP**4**, and SP**5** may be referred to as hexadeca cells.

[0069] A shared pixel may include sub-pixels that are adjacent to each other while including a same color filter. A shared pixel of FIGS. **3**A to **3**C is illustrated as an example including sub-pixels having an N*N arrangement, but an arrangement of sub-pixels included in a shared pixel is not limited to N*N. N may be a natural number equal to or greater than 2.

[0070] FIG. **4**A is a diagram for describing a circuit diagram of a hybrid pixel PXa, according to some example embodiments.

[0071] Referring to FIG. **4**A, the hybrid pixel PXa may include a photodiode **3000**, a pixel circuit **1000** connected to one end of the photodiode **3000**, and a sensing circuit **2000** connected to another end of the photodiode **3000**. According to some example embodiments, a cathode of the photodiode **3000** may be connected to the pixel circuit **1000** and an anode of the photodiode **3000** may be connected to the sensing circuit **2000**.

[0072] The pixel circuit **1000** may generate a pixel signal of a voltage according to an amount of charged generated in the photodiode **3000**. The sensing circuit **2000** may generate an event detection signal by detecting whether an amount of change in charges generated in the photodiode **3000** has exceeded a certain threshold value. The pixel circuit **1000** included in the hybrid pixel PXa according to the inventive concepts may generate the pixel signal based on electrons generated in the photodiode **3000**, and the sensing circuit **2000** may generate the event detection signal based on holes generated in the photodiode **3000**. Accordingly, the pixel circuit **1000** may use a current based on electrons and the sensing circuit **2000** may use a current based on holes. In this case, even when electrons are accumulated in the photodiode **3000**, the current based on holes may continuously flow, and thus, the pixel circuit **1000** may maintain a 4T operation, operate as a circuit separate from the sensing circuit **2000**, and may have various share structures. In some example embodiments, the pixel signal generated from electrons may have less noise compared to when a pixel signal is generated from holes, and as such, a signal quality of a pixel signal may be improved. In some example embodiments, the sensing circuit **2000** uses a hole current, and thus, the hybrid sensor PXa may be implemented with improved performance and reliability with regards to light loss or pixel operation.

[0073] FIG. **4**B illustrates some example embodiments of a circuit diagram of the hybrid pixel PXa of FIG. **4**A.

[0074] Referring to FIG. **4**B, the hybrid pixel PXa may include a photodiode **3000***a*, a pixel circuit **1000***a*, and a sensing circuit **2000***a*. The pixel circuit **1000***a* may include a transfer transistor TX, a reset transistor RX, a driving transistor DX, and a selection transistor SX. Description about an operating method of the pixel circuit **1000***a* is omitted. Referring to FIG. **4**B, one end of the transfer transistor TX included in the pixel circuit **1000***a* may be connected to a cathode of the photodiode **3000***a*. The sensing circuit **2000***a* may include a transimpedance amplifier TIA. According to some example embodiments, the transimpedance amplifier TIA may amplify a value of current corresponding to holes output from an anode of the photodiode **3000***a* and output the same (or about the same) as a voltage. Although omitted for convenience of description in the inventive concepts, the sensing circuit **2000***a* may further include a comparator configured to compare an amplified hole current value with a reference value. According to some example embodiments, the sensing circuit **2000***a* may be a circuit corresponding to a dynamic vision sensor configured to output an event detection signal. Description about an operating method of the sensing circuit **2000***a* will be omitted.

[0075] Circuit structures of the pixel circuit **1000***a* and sensing circuit **2000***a* of FIG. **4**B are only examples, and the pixel circuit **1000***a* and the sensing circuit **2000***a* may be variously changed respectively within ranges capable of performing a circuit structure of reading out a pixel signal and a circuit structure of comparing an amount of change in charges with a threshold value and outputting an event detection signal. For example, according to some example embodiments, there may be an increase in speed, accuracy, and/or power efficiency of the image processing device based on the above methods. Therefore, the improved devices and methods overcome the deficiencies of the conventional devices and methods while reducing resource consumption, and improving data accuracy, and resource allocation (e.g., latency).

[0076] According to some example embodiments, the pixel circuit **1000***a* and the sensing circuit **2000***a* included in the hybrid pixel PXa may share the photodiode **3000***a*. Accordingly, both a function of an image sensor and a function of a dynamic vision sensor may be performed based at least one photodiode **3000***a* included in one hybrid pixel PXa.

[0077] FIG. **5**A is a cross-sectional view of a layout region corresponding to an A region of FIG. **4**B. Referring to FIG. **5**A, the hybrid pixel PXa may include a semiconductor substrate **2110**, a photoelectric conversion area **2120**, a vertical transfer gate VTG, a micro-lens **2130**, a color filter **2140**, deep trench isolation (DTI) areas **2150** and **2160**, and a contact area **2180**.
[0078] The semiconductor substrate **2110** may include a first surface SUF**1** and a second surface SUF**2** facing the first surface SUF**1**. Impurities (e.g., boron (B)) of a first conductivity type (e.g., a p type) may be injected into the semiconductor substrate **2110**, and impurities (e.g., phosphorus (P) or arsenic (As)) of a second conductivity type (e.g., an n type) opposite to the first conductivity type may be injected to a floating diffusion area. Although not illustrated, a p-type epitaxial layer may be grown or a separate well area may be formed above the semiconductor substrate **2110** to form the photoelectric conversion area **2120** and the vertical transfer gate VTG on the p-type epitaxial layer and/or the well area.
[0079] The photoelectric conversion area **2120** may be formed inside the semiconductor substrate **2110** and incident light may reach the photoelectric conversion area **2120** through the second surface SUF**2** of the semiconductor substrate **2110**. The photoelectric conversion area **2120** may be an area corresponding to the photodiode **3000**. The photoelectric conversion area **2120** may generate charges (e.g., photo charges or holes) in a PN junction region according to photoelectric conversion, based on incident light. The number of electrons generated may increase as luminance increases. The number of holes generated may be the same as the number of electrons. According to some example embodiments, the photoelectric conversion area **2120** may be an area formed as the second conductivity type (e.g., the n type) is injected. An area and shape of the photoelectric conversion area **2120** are not limited to those shown in FIG. **5**A.
[0080] Referring to FIG. **5**A, the semiconductor substrate **2110** may include a first area **2120***a* and a second area **2120***b*. According to some example embodiments, the first area **2120***a* may be an area corresponding to the photoelectric conversion area **2120**. According to some example embodiments, the second area **2120***b* may be an area surrounding the first area **2120***a* in the semiconductor substrate **2110**. According to some example embodiments, the second area **2120***b* may be a remaining area obtained by excluding an area where the photoelectric conversion area **2120** from the semiconductor substrate **2110**. According to some example embodiments, the second area **2120***b* may be an area in the semiconductor substrate **2110**, where hole transfer is possible between areas where holes are accumulated.
[0081] According to some example embodiments, the contact area **2180** may be formed on the first surface SUF**1** of the semiconductor substrate **2110**. According to some example embodiments, the contact area **2180** may be formed in the semiconductor substrate **2110** and provide a transfer path of holes formed in the second area **2120***b*. According to some example embodiments, holes formed in the second area **2120***b* may be transferred to a sensing circuit **2000***a* along the contact area **2180**.
[0082] The vertical transfer gate VTG may be formed on the first surface SUF**1** of the semiconductor substrate **2110**. A portion of the vertical transfer gate VTG may extend from the first surface SUF**1** of the semiconductor substrate **2110** into the semiconductor substrate **2110** towards the photoelectric conversion area **2120**. The vertical transfer gate VTG may allow electrons generated in the first area **2120***a* of the photoelectric conversion area **2120** to be transferred to a floating diffusion area.
[0083] According to the inventive concepts, electrons generated in the first area **2120***a* may be transferred to the pixel circuit **1000***a* through the vertical transfer gate VTG, and holes generated in the second area **2120***b* may be transferred to the sensing circuit **2000***a* through the contact area **2180**.
[0084] The hybrid pixel PXa may include the DTI areas **2150** and **2160**. According to some example embodiments, the DTI areas **2150** and **2160** may extend from the first surface SUF**1** of the semiconductor substrate **2110** to a depth spaced apart from the first surface SUF**1**, or may completely (e.g., fully) penetrate the semiconductor substrate **2110** from the first surface SUF**1** to

the second surface SUF**2** of the semiconductor substrate **2110**. The DTI areas **2150** and **2160** may be in contact with the first surface SUF**1** and/or the second surface SUF**2** of the semiconductor substrate **2110**. According to some example embodiments, surfaces where the DTI areas **2150** and **2160** and the semiconductor substrate **2110** are in contact with each other may not be parallel to each other. According to some example embodiments, the DTI areas **2150** and **2160** may from isolation areas by contacting a shallow trench isolation (STI) area formed in the first surface SUF**1** or the second surface SUF**2**. According to some example embodiments, the DTI areas **2150** and **2160** may be isolation areas formed by front DTI (FDTI). The DTI areas **2150** and **2160** may include an insulating layer and/or a conductive layer. For example, the DTI areas **2150** and **2160** may include a silicon oxide layer formed along an inner wall of a trench and a silicon layer filling a remaining portion of the trench. According to some example embodiments, thicknesses of the DTI areas **2150** and **2160** may not be uniform.

[0085] According to the inventive concepts, the DTI areas **2150** and **2160** may be formed to isolate the first area **2120***a* and the second area **2120***b* included in the hybrid pixel PXa from a first area and a second area included in an adjacent hybrid pixel Pxa. According to the inventive concepts, the first area **2120***a* and the second area **2120***b* are formed inside the semiconductor substrate **2110** and the contact area **2180** is formed above the semiconductor substrate **2110**, and to secure a complete transfer path of holes, the second area **2120***b* and the contact area **2180** corresponding thereto need to be completely (e.g., fully) isolated from a contact area arranged in an adjacent pixel. Accordingly, the DTI areas **2150** and **2160** may be formed on both sides of the first area **2120***a* and the second area **2120***b* to configure an individual pixel for the sensing circuit **2000***a* (and, for example, other sides not shown). According to the inventive concepts, the sensing circuit **2000***a* may be configured by using hole currents classified for each hybrid pixel by the DTI areas **2150** and **2160**.

[0086] The hybrid pixel PXa may further include the color filter array CF and the micro-lens **2130** on the second surface SUF**2** of the semiconductor substrate **2110**. In some example embodiments, the order of the color filter array CF and the micro-lens **2130** may be different. According to some example embodiments, a nano-structure may be located on the second surface SUF**2** of the semiconductor substrate **2110**, instead of the color filter array CF and/or the micro-lens **2130**, and may isolate and/or guide light according to wavelengths.

[0087] FIG. **5**B is a diagram for describing a flow of charges in the cross-sectional view of FIG. **5**A. FIG. **5**B is a diagram for describing a flow of charges from the first area **2120***a* and the second area **2120***b* included in the photoelectric conversion area **2120**.

[0088] Referring to FIG. **5**B, light L transmitted through the micro-lens **2130** may be transmitted to the photoelectric conversion area **2120**. When a voltage is applied to the vertical transfer gate VTG, electrons e-formed in the first area **2120***a* by the light L may flow in a direction towards the vertical transfer gate VTG and may be transmitted to the floating diffusion area of the pixel circuit **1000***a*. Holes h+ formed in the second area **2120***b* by the light L may be transferred through the contact area **2180** and a hole current transferred through the contact area **2180** may be transmitted to the sensing circuit **2000***a*.

[0089] According to some example embodiments, electrons may be used to generate a pixel signal and holes may be used to detect an event. Compared to a case where only electrons are used according to a comparative example, a level of a pixel signal may be increased under a same or similar luminance, thereby, a precision of detecting an event may be increased, and image quality at a dark place where luminance is low may be increased.

[0090] FIG. **6**A is a diagram for describing a circuit diagram of a hybrid pixel PXb, according to some example embodiments.

[0091] Referring to FIG. **6**A, the hybrid pixel PXb may include a pixel circuit **1000***b*, a sensing circuit **2000***b*, and a plurality of photodiodes **3000***b*. The pixel circuit **1000***b* according to some example embodiments may include first to fourth transfer transistors TX**1**, TX**2**, TX**3**, and TX**4**, a first floating diffusion node FD**1**, a source follower transistor SF**1**, a first selection transistor SX**1**, and first and second reset transistors RX**1** and RX**2**. An operating method of the pixel circuit **1000***b* may be similar to that of the pixel circuit **1000**, and thus, description thereof is omitted.

[0092] Referring to FIG. **6**A, the plurality of photodiodes **3000***b* may include first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4**. Charges corresponding to light incident on the hybrid pixel PXb through the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** may be accumulated. An amount of charges accumulated in the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** may be referred to as light amount data.

[0093] The pixel circuit **1000***b* may be connected to cathodes of the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4**. The sensing circuit **2000***b* may be connected to anodes of the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4**. According to some example embodiments, the pixel circuit **1000***b* and the sensing circuit **2000***b* may share the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4**. In some example embodiments of FIG. **6**A, the pixel circuit **1000***b* and the sensing circuit **2000***b* share four photodiodes, but the inventive concepts are not limited thereto, and the pixel circuit **1000***b* and the sensing circuit **2000***b* may share N photodiodes. N may be a natural number equal to or greater than 2. According to some example embodiments, the number of transfer transistors included in the pixel circuit **1000***b* and the number of photodiodes **3000***b* connected to the pixel circuit **1000***b* may be the same. Referring to FIG. **6**A, the number of transfer transistors included in the pixel circuit **1000***b* may be 4 and the number of photodiodes **3000***b* connected to the pixel circuit **1000***b* may be 4.

[0094] FIG. **6**B is a layout plan view illustrating some components of the hybrid pixel PXb of FIG. **6**A. FIG. **6**C is a cross-sectional view taken alon′ a line A-A′ of FIG. **6**B.

[0095] Referring to FIG. **6**B, four photodiodes, e.g., the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4**, included in FIG. **6**A may be provided in a 2×2 pixel structure symmetrical based on the center.

[0096] The hybrid pixel PXb of FIG. **6**B may include one semiconductor substrate **510**, a plurality of photoelectric conversion areas **520**, one floating diffusion area **530**, four vertical transfer gates **540**, a DTI area **560**, a transistor **590**, and a contact area **591**. The hybrid pixel PXb of FIG. **6**B may have a 4PD structure in which the semiconductor substrate **510** is shared as a center portion of the DTI area **560** is partially deleted.

[0097] According to some example embodiments of FIG. **6**B, the plurality of photoelectric conversion areas **520** may include first areas **520***a*, **520***b*, **520***c*, and **520***d*. According to some example embodiments, the first areas **520***a*, **520***b*, **520***c*, and **520***d* may be areas doped with a second conductivity type (e.g., an n type). Referring to FIG. **6**B, the number of first areas **520***a*, **520***b*, **520***c*, and **520***d* may correspond to the number of photodiodes shared by the pixel circuit **1000***b*. Referring to FIG. **6**B, there are four photodiodes shared by the pixel circuit **1000***b*, and thus, the number of first areas **520***a*, **520***b*, **520***c*, and **520***d* may be four. According to some example embodiments of FIG. **6**B, the semiconductor substrate **510** may include a second area **520***c* surrounding the first areas **520***a*, **520***b*, **520***c*, and **520***d*. According to some example embodiments, the second area **520***c* surrounding the plurality of first areas **520***a*, **520***b*, **520***c*, and **520***d* is illustrated. Also, the contact area **591** may be formed above the semiconductor substrate **510**. According to some example embodiments, one contact area **591** may correspond to one second area **520***c*. In other words, the number of second areas **520***c* and the number of contact areas **591** may be the same. Referring to FIG. **6**B, the plurality of first areas **520***a*, **520***b*, **520***c*, and **520***d* share the second area **520***e* and share the contact area **591**, and thus, the number of contact areas **591** and the number of second areas **520***e* may be equal to or less than the number of first areas **520***a*, **520***b*, **520***c*, and **520***d*. A location where the contact area **591** is arranged may vary within a range in contact with the second area **520***c*, above the semiconductor substrate **510**.

[0098] Referring to FIG. **6**C, the hybrid pixel PXb includes the semiconductor substrate **510**, the first areas **520***c* and **520***d*, the second area **520***c*, the contact area **591**, the vertical transfer gate **540**, the color filter **2140**, the micro-lens **2130**, and the DTI area **560**. Details about the semiconductor substrate **510**, the vertical transfer gate **540**, the color filter **2140**, and the micro-lens **2130** correspond to those about respective components described with reference to FIG. **5**A, and thus, redundant description will be omitted.

[0099] Referring to FIG. **6**C, the second area **520***e* is formed within a range surrounding the first areas **520***c* and **520***d*, and the contact area **591** is formed in the second area **520***c*, and thus, holes generated in the second area **520***e* may be transferred to a sensing circuit. According to some example embodiments, the first areas **520***c* and **520***d* may share the second area **520***c*.

[0100] According to the inventive concepts, in a hybrid pixel sharing 4PD, one second area **520***e* may be formed to surround four first areas **520***a*, **520***b*, **520***c*, and **520***d*, and thus, the second area **520***e* may be shared. According to some example embodiments, it should be noted that, in the hybrid pixel sharing 4PD, the four first areas **520***a*, **520***b*, **520***c*, and **520***d* are considered as one pixel, and thus, the DTI area **560** may be formed for distinguishment from an adjacent hybrid pixel.

[0101] In comparison with FIG. **5**A, when the first area **2120***a* and the second area **2120***b*, which are included in the photoelectric conversion area **2120**, are in a 1:1 correspondence in FIG. **5**A, the DTI areas **2150** and **2160** may be formed such that the photoelectric conversion area **2120** is isolated from an adjacent photoelectric conversion area. In FIG. **6**C, when the first areas **520***a*, **520***b*, **520***c*, and **520***d* and the second area **520***e*, which are included in the photoelectric conversion areas **520**, are not in a 1:1 correspondence, the plurality of first areas **520***a*, **520***b*, **520***c*, and **520***d* may share the second area **520***c*. In this case, the DTI area **560** may be formed such that the second area **520***e* is isolated from an adjacent second area. In other words, it should be noted that a DTI area may be formed such that a second area of a photoelectric conversion area in a same conductivity type as a substrate is completely (e.g., fully) isolated from a second area of an adjacent photoelectric conversion area.

[0102] FIG. **7** is a circuit diagram of a hybrid pixel PXc according to some example embodiments.

[0103] Referring to FIG. **7**, the hybrid pixel PXc may include a photodiode **3000***c*, a pixel circuit **1000***c* connected to one end of the photodiode **3000***c*, and a sensing circuit **2000***c* connected to another end of the photodiode **3000***c*. According to some example embodiments, an anode of the photodiode **3000***c* may be connected to the pixel circuit **1000***c* and a cathode of the photodiode **3000***c* may be connected to the sensing circuit **2000***c*.

[0104] The pixel circuit **1000***c* may generate a pixel signal of a voltage according to an amount of charged generated in the photodiode **3000***c*. The sensing circuit **2000***c* may generate an event detection signal by detecting whether an amount of change in charges generated in the photodiode **3000***c* has exceeded a certain threshold value. The pixel circuit **1000***c* included in the hybrid pixel PXc according to the inventive concepts may generate the pixel signal based on holes generated in the photodiode **3000***c*, and the sensing circuit **2000***c* may generate the event detection signal based on electrons generated in the photodiode **3000***c*. Accordingly, the pixel circuit **1000***c* may use a current based on holes and the sensing circuit **2000***c* may use a current based on electrons. In this case as well, the pixel circuit **1000***c* may operate as a circuit separate from the sensing circuit **2000***c*.

[0105] It should be noted that, the inventive concepts as applied to some example embodiments of a pixel circuit using electrons and a sensing circuit using holes may also be applied to a pixel circuit using holes and a sensing circuit using electrons, as in FIG. **7**.

[0106] FIG. **8** illustrates a stack structure of a hybrid sensor **1**, according to some example embodiments.

[0107] Referring to FIG. **8**, the hybrid sensor **1** may include an upper chip **40** and a lower chip **50**. The upper chip **40** may include a sensing area SA where some circuits of the plurality of pixels PX are provided, a circuit area LC where elements for driving the plurality of pixels PX are provided, and pad areas PA**1** and PA**2** around the sensing area SA and the circuit area LC. A plurality of upper pads PAD are arranged in the pad areas PA**1** and PA**2** and the plurality of upper pads PAD may be connected to elements provided in the lower chip **50** through a via and/or the like.

[0108] The lower chip **50** includes the circuit area LC, and peripheral circuits of a pixel array, e.g., the row driver **120**, the readout circuit **130**, the ramp signal generator **140**, the timing controller **150**, and the event detection circuit **160**, may be formed in the circuit area LC. According to some example embodiments, a sensing circuit included in the plurality of pixels PX may be formed in the circuit area LC. This will be described below with reference to FIGS. **10**A and **10**B. According to some example embodiments, the lower chip **50** may include a memory area and a dummy area. Memory elements, such as dynamic random access memory (DRAM) elements or static random access memory (SRAM) elements, may be arranged in the memory area. However, memory elements arranged in the memory area are not limited to DRAM elements or SRAM elements. The dummy area may perform a function, for example, of supporting the upper chip **40**, heat dissipation, etc., instead of a function of storing data.

[0109] FIG. **9** illustrates a stack structure of a hybrid sensor, according to some example embodiments.

[0110] Referring to FIG. **9**, the hybrid sensor **2** may include a plurality of stacked chips. For example, a pixel array may be formed in the upper chip **40** and an intermediate chip **51**, and peripheral circuits of the pixel array or a memory may be formed in a lower chip **60**. According to some example embodiments, a plurality of photodiodes, a pixel circuit, and wires corresponding thereto, which are included in a hybrid pixel, may be formed in the upper chip **40**, and a sensing circuit and wires corresponding thereto may be formed in the intermediate chip **51**.

[0111] The lower chip **60** may include the circuit area LC and the peripheral circuits of the pixel array may be formed in the circuit area LC. According to some example embodiments, the lower chip **60** may include a memory area and a dummy area.

[0112] According to some example embodiments, the upper chip **40** and the intermediate chip **51** may be stacked on each other in a wafer level, and the lower chip **60** may be attached below the intermediate chip **51** in a chip level.

[0113] FIG. **10**A illustrates a stack structure of components of a hybrid pixel included in a hybrid sensor, according to some example embodiments.

[0114] For convenience of description, some example embodiments in which a wiring structure of the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4**, a pixel circuit, and a sensing circuit, which are included in one hybrid pixel **11**, is applied to a stack structure is illustrated.

[0115] Referring to FIG. **10**A, the hybrid pixel **11** may include an upper chip **41** and a lower chip **42**. The upper chip **41** may include a first area **41***a* where the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** are formed, and a second area **41***b* where a pixel circuit and wires corresponding thereto are formed, below the first area **41***a*. A sensing circuit and wires corresponding thereto may be formed in the lower chip **42**. An object lens **46** for applying light to the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** included in the upper chip **41** may be arranged on an upper area of the upper chip **41**.

[0116] According to some example embodiments, the upper chip **41** of the hybrid pixel **11** may correspond to the upper chip **40** of FIG. **8**, and the lower chip **42** of the hybrid pixel **11** may correspond to the lower chip **50** of FIG. **8**. According to some example embodiments, the upper chip **41** of the hybrid pixel **11** may correspond to the upper chip **40** of FIG. **9**, and the lower chip **42** of the hybrid pixel **11** may correspond to the intermediate chip **51** of FIG. **9**.

[0117] According to some example embodiments, the pixel circuit uses electrons and thus may include an n-channel metal oxide semiconductor (NMOS) transistor, and the sensing circuit may include both an NMOS transistor and a p-channel metal oxide semiconductor (PMOS) transistor. Thus, an efficient stack structure may be formed by forming the sensing circuit including both the NMOS transistor and the PMOS transistor on a layer separate from the pixel circuit and a photodiode.

[0118] According to some example embodiments, the plurality of first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** may be shared with each other and correspond to one hybrid pixel. According to some example embodiments, the hybrid pixel **11** including the plurality of first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** may be isolated from an adjacent hybrid pixel through a DTI area **41***c*. According to some example embodiments, the DTI area **41***c* may be formed according to a depth of the first area **41***a* where the plurality of first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** are formed. Referring to FIG. **10**A, the DTI area **41***c* is formed only on a right side of the first area **41***a* where the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** are formed, but this is for convenience of description, and it should be noted that the DTI area **41***c* is formed to surround all four sides of the first area **41***a* where the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** are formed. According to the inventive concepts, by using the DTI area **41***c* that is physically separated from an adjacent hybrid pixel, individual sensing pixels respectively corresponding to sensing circuits may be formed.

[0119] Although not illustrated in FIG. **10**A, the plurality of first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** may include a first area and a second area, and may share the second area. Details thereof correspond to those described with reference to FIGS. **6**A to **6**C, and thus, redundant description will be omitted.

[0120] According to some example embodiments, the upper chip **41** and the lower chip **42** may be electrically connected to each other through a vertical area **45**. The vertical area **45** extends in a Z-axis direction and may be in contact with each of the upper chip **41** and the lower chip **42**. According to some example embodiments, the vertical area **45** may be a via for connecting charges generated in the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** to the sensing circuit. According to some example embodiments, the vertical area **45** may be a through silicon via (TSV) or a chip-to-chip (C2C) for connecting the charges of the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** to the sensing circuit.

[0121] FIG. **10**B illustrates a stack structure of components of a hybrid pixel **21** included in a hybrid sensor, according to some example embodiments.

[0122] For convenience of description, some example embodiments in which a wiring structure of photodiodes, a pixel circuit, and a sensing circuit, which are included in a plurality of hybrid pixels, is applied to a stack structure is illustrated. According to some example embodiments, an upper chip **61** of the hybrid pixel **21** may correspond to the upper chip **40** of FIG. **8**, and a lower chip **62** of the hybrid pixel **21** may correspond to the lower chip **50** of FIG. **8**. According to some example embodiments, the upper chip **61** of the hybrid pixel **21** may correspond to the upper chip **40** of FIG. **9**, and the lower chip **62** of the hybrid pixel **21** may correspond to the intermediate chip **51** of FIG. **9**.

[0123] Referring to FIG. **10**B, the hybrid pixel **21** may include the upper chip **61** and the lower chip **62**. The upper chip **61** may include a first area **61***a* where the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** are formed, and a second area **61***b* where a pixel circuit and wires corresponding thereto are formed, below the first area **61***a*. A sensing circuit and wires corresponding thereto may be formed in the lower chip **62**. Object lenses **66** for applying light to the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** included in the upper chip **61** may be arranged on an upper area of the upper chip **61**.

[0124] FIG. **10**B illustrates a stack structure in which sixteen hybrid pixels **11** of FIG. **10**A are arranged. A B region of FIG. **10**B corresponds to the first area **41***a* and the second area **41***b* formed in the upper chip **41** of FIG. **10**A, and thus, redundant description thereof is omitted.

[0125] Referring to FIG. **10**B, a plurality of hybrid pixels may be physically separated by DTI areas **61***c*. In some example embodiments of FIGS. **10**A and **10**B, a hybrid pixel may denote a pixel corresponding to one unit sharing the plurality of first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4**. According to some example embodiments, the DTI area **61***c* may be formed according to a depth of the first area **61***a* where the plurality of first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** are formed. Referring to FIG. **10**B, some example embodiments in which the DTI areas **61***c* are formed only between four hybrid pixels in front of the first area **61***a* where the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** are formed is illustrated, but this is only for convenience of description, and it should be noted that the DTI area **61***c* is formed to surround each of hybrid pixels sharing the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4**, in order to prevent or reduce crosstalk with an adjacent hybrid pixel.

[0126] Although not illustrated in FIG. **10**B, the plurality of first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** may include a first area and a second area, and may share the second area. Details thereof correspond to those described with reference to FIGS. **6**A to **6**C, and thus, redundant description will be omitted.

[0127] Referring to FIG. **10**B, the upper chip **61** and the lower chip **62** may be electrically connected to each other through a vertical area **65**. The vertical area **65** extends in a Z-axis direction and may be in contact with each of the upper chip **61** and the lower chip **62**. According to some example embodiments, the vertical area **65** may be a via for connecting charges of the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** to a sensing circuit. According to some example embodiments, the vertical area **65** may be a TSV or C2C for connecting charges of the first to fourth photodiodes PD**1**, PD**2**, PD**3**, and PD**4** to the sensing circuit.

[0128] Referring to FIG. **10**B, the vertical areas **65** for connecting holes of a plurality of hybrid pixels to one sensing circuit may be provided instead of providing one vertical area **45** for connecting a pixel circuit and a sensing circuit, included in one hybrid pixel, as shown in FIG. **10**A.

[0129] Referring to FIG. **10**B, to connect outputs of photodiodes included in each of a plurality of hybrid pixels to one, a plurality of first vertical areas **65***b* perpendicularly connected to an output of each hybrid pixel (for example, electrically connected), a plurality of first planar areas **65***c* connecting the plurality of first vertical areas **65***b* on an X-Y plane (for example, electrically connected), and at least one second vertical area **65***a* connecting the plurality of first vertical areas **65***b* and the plurality of first planar areas **65***c* to the sensing circuit (for example, electrically connected) may be provided. According to some example embodiments, each of the plurality of first vertical areas **65***b* may be vertical areas for connecting charges respectively corresponding to contact areas included in hybrid pixels, that is, each of the plurality of first vertical areas **65***b* may be electrically connected to corresponding contact areas included in the hybrid pixels.

[0130] Referring to FIG. **10**B, only one second vertical area **65***a* is illustrated, but this is only an example, and it should be noted that a plurality of second vertical areas **65***a* may be provided. Referring to FIG. **10**B, the contact areas separated by DTI areas in the upper chip **61** may be connected to one sensing circuit by being combined into one wire through wires such as metal or polysilicon included in the vertical areas **65**.

[0131] Referring to some example embodiments of FIG. **10**B, when the sensing circuit performs event detection by mixing outputs of sixteen hybrid pixels, it is possible to connect 16 times greater signals, and accordingly, operation in low illumination may be facilitated. According to some example embodiments, outputs of hole signals of a plurality of hybrid pixels may be connected as shown in FIG. **10**B to secure the area and sensitivity of a sensing circuit. According to some example embodiments, when color information is not required in a sensing circuit, a wire may be formed by connecting hole signals of photodiodes corresponding to RGB.

[0132] FIG. **11** is a block diagram of a structure of a hybrid pixel PXd, according to some example embodiments.

[0133] Referring to FIG. **11**, the hybrid pixel PXd may include a photodiode **3000***d*, a pixel circuit **1000***d*, a sensing circuit **2000***d*, and a noise removal circuit **4000***d*. While describing FIG. **11**, details that overlap those described in FIG. **4**A will be omitted. According to some example embodiments, in the sensing circuit **2000***d* using holes, noise caused by a TIA may be occur and when the noise shakes a ground level, signal-to-noise ratio (SNR) of a pixel signal may be affected. Referring to FIG. **11**, the noise removal circuit **4000***d* may be connected between the photodiode **3000***d* and the sensing circuit **2000***d*. The noise removal circuit **4000***d* may be connected between

an anode of the photodiode **3000***d* and the sensing circuit **2000***d*. The noise removal circuit **4000***d* may include a plurality of transistors and noise occurred by the sensing circuit **2000***d* may be removed by adjusting on-off of the plurality of transistors.
[0134] FIG. **12** is a circuit diagram of a hybrid pixel PXe, according to some example embodiments.
[0135] Referring to FIG. **12**, the hybrid pixel PXe may include a pixel circuit **1000***c*, a sensing circuit **2000***c*, a photodiode **3000***c*, and a noise removal circuit **4000***c*. The pixel circuit **1000***c* may include the transfer transistor TX, the reset transistor RX, the driving transistor DX, and the selection transistor SX. A structure of the sensing circuit **2000***e* may correspond to a structure of the sensing circuit **2000** shown in FIG. **4**A. According to some example embodiments, details about the pixel circuit **1000***e*, the sensing circuit **2000***c*, and the photodiode **3000***e* correspond to details about a pixel circuit, a sensing circuit, and a photodiode described with reference to FIGS. **4**A to **10**B, and thus, redundant description will be omitted.
[0136] The noise removal circuit **4000***c* may include a first transistor NRB and a second transistor NR. According to some example embodiments, the first transistor NRB may be connected between an anode of the photodiode **3000***e* and the sensing circuit **2000***c*. The second transistor NR may be connected between the anode of the photodiode **3000***e* and a ground. According to some example embodiments, the first transistor NRB and the second transistor NR may be connected to each other in parallel. The noise removal circuit **4000***e* may configure a circuit connecting or releasing the sensing circuit **2000***e* by using the first transistor NRB and the second transistor NR. According to some example embodiments, the first transistor NRB of the noise removal circuit **4000***c* may be a transistor configured to control a connection between a contact area of the photodiode **3000***e* and the sensing circuit **2000***e*, and the second transistor NR may be a transistor configured to control a connection between the contact area of the photodiode **3000***e* and the ground or between different specific voltages.
[0137] When the first transistor NRB is turned on and the second transistor NR is turned off, the photodiode **3000***e* and the sensing circuit **2000***e* may be connected to each other. In this case, holes generated in the photodiode **3000***e* may be transferred to the sensing circuit **2000***c*. When the first transistor NRB is turned off and the second transistor NR is turned on, the photodiode **3000***e* and the ground may be connected to each other and the holes generated in the photodiode **3000***c* may flow to the ground, and thus, a hole current may be drained. As such, by controlling on and off of the first transistor NRB and the second transistor NR, a connection between the sensing circuit **2000***e* and the photodiode **3000***e* may be adjusted.
[0138] FIG. **13** is a timing diagram for describing turn-on timing of transistors included in the noise removal circuit **4000***e* of FIG. **12**.
[0139] FIG. **13** illustrates a timing diagram of a selection control signal SEL applied to the selection transistor SX, a reset control signal RG applied to the reset transistor RX, a transfer control signal TG applied to the transfer transistor TX, a first control signal NRBS applied to the first transistor NRB, and a second control signal NRS applied to the second transistor NR.
[0140] Referring to FIG. **13**, sections corresponding to one frame is illustrated. During one frame FRM, a reset time RST, an exposure time IT, a read time RO, and a non-integration time NIT may be assigned for each of a plurality of rows. The reset time RST corresponding to sections of t**1** to t**3**, the exposure time IT corresponding to sections of t**3** and t**4**, the read time RO corresponding to sections of t**4** to **16**, and the non-integration time NIT corresponding to sections of t**6** and t**7** are illustrated.
[0141] During the reset time RST corresponding to the sections of t**1** to t**3**, a pixel may be reset. At a time point t**2**, the transfer transistor TX included in the pixel may be turned on and transmit charges generated in a photodiode during the non-integration time NIT to a floating diffusion node, thereby removing the charges. During the reset time RST, the reset control signal RG applied to the reset transistor RX maintains a second level, and thus, the transfer transistor TX of the pixel is

turned on while a reset voltage is applied to the floating diffusion node. Accordingly, the floating diffusion node and the pixel may be reset together.

[0142] During the exposure time IT corresponding to the sections t**3** and t**4**, charges corresponding to an optical signal may be generated and accumulated in a photodiode included in the pixel. The pixel may be read during the read time RO corresponding to the sections t**4** to **16**. At a time point t**5**, the transfer transistor TX included in the pixel is turned on and the charges accumulated in the photodiode are transmitted to the floating diffusion node during the exposure time IT, and a pixel voltage corresponding to the transmitted charges may be output through the column line CL of FIG. **1**. During the read time RO before the time point t**5**, sampling (reset sampling) of a signal corresponding to the reset voltage may be performed, and during the read time RO after the time point t**5**, sampling (signal sampling) of a signal corresponding to an image voltage may be performed. At a time point t**4** that is a start time point of the read time RO, the selection control signal SEL applied to the selection transistor SX may be changed from a first level to the second level, and the reset control signal RG applied to the reset transistor RX may be changed from the second level to the first level. At a time point t**6** that is an end time point of the read time RO, the selection control signal SEL applied to the selection transistor SX may be changed from the second level to the first level, and the reset control signal RG applied to the reset transistor RX may be changed from the first level to the second level. In the inventive concepts, the first level may be a low level and the second level may be a high level.

[0143] According to some example embodiments, during the reset time RST, the exposure time IT, and the non-integration time NIT, the second control signal NRS applied to the second transistor NR may maintain the first level, and during the read time RO, the second control signal NRS applied to the second transistor NR may maintain the second level. According to some example embodiments, during the reset time RST, the exposure time IT, and the non-integration time NIT, the first control signal NRBS applied to the first transistor NRB may maintain the second level, and during the read time RO, the first control signal NRBS applied to the first transistor NRB may maintain the first level. According to some example embodiments, at the time point t**4** that is the start time point of the read time RO, the second control signal NRS may be changed from the first level to the second level, and the first control signal NRBS may be changed from the second level to the first level. According to some example embodiments, at the time point t**6** that is the end time point of the read time RO, the second control signal NRS may be changed from the second level to the first level, and the first control signal NRBS may be changed from the first level to the second level. According to some example embodiments, the first transistor NRB and the second transistor NR may be complementary. The second transistor NR may be turned off when the first transistor NRB is turned on, and the second transistor NR may be turned on when the first transistor NRB is turned off.

[0144] According to some example embodiments, the first control signal NRBS applied to the first transistor NRB may be changed from the second level to the first level before a CDS sampling time ST, and may be changed from the first level to the second level after the CDS sampling time ST. According to some example embodiments, the second control signal NRS applied to the second transistor NR may be changed from the first level to the second level before the CDS sampling time ST, and may be changed from the second level to the first level after the CDS sampling time ST. The CDS sampling time ST may be a section included in the read time RO.

[0145] In other words, the first transistor NRB is turned off and the second transistor NR is turned on during the read time RO so as to block a path that may be connected to the sensing circuit **2000***e* during the read time RO and block noise that may occur by the sensing circuit **2000***e* during the read time RO. According to some example embodiments, the read time RO may be most vulnerable to noise, and thus, noise may be removed by controlling signals applied to the transistors included in the noise removal circuit **4000***e* according to the inventive concepts.

[0146] FIGS. **14** and **15** are timing diagrams for describing turn-on timing of transistors included in

a noise removal circuit according to some example embodiments. While describing FIGS. **14** and **15**, details that overlap those described with reference to FIG. **13** will be omitted.
[0147] Referring to FIG. **14**, during the reset time RST corresponding to the sections of t**1** to t**3**, the second control signal NRS applied to the second transistor NR may maintain the second level and the first control signal NRBS applied to the first transistor NRB may maintain the first level. According to some example embodiments, at a time point t**1** that is a start time point of the reset time RST, the second control signal NRS may be changed from the first level to the second level, and the first control signal NRBS may be changed from the second level to the first level. According to some example embodiments, at a time point t**3** that is an end time point of the reset time RST, the second control signal NRS may be changed from the second level to the first level, and the first control signal NRBS may be changed from the first level to the second level.
[0148] Referring to some example embodiments of FIG. **14**, by controlling the second transistor NR to be turned on and the first transistor NRB to be turned off during the reset time RST corresponding to the sections of t**1** to t**3**, noise that may occur in the sensing circuit **2000***c* may have little to no influence even during a reset operation.
[0149] In FIGS. **13** and **14**, some example embodiments in which the second transistor NR is turned on and the first transistor NRB is turned off during the entire section of the reset time RST and/or the read time RO is illustrated, but the inventive concepts are not limited thereto, and the second transistor NR may be turned on and the first transistor NRB may be turned off during a time corresponding to 90% or more of the entire section of the reset time RST and/or the read time RO. According to some example embodiments, the second transistor NR may be turned off and the first transistor NRB may be turned on in a partial section of the reset time RST and/or the read time RO. According to some example embodiments, during the reset time RST, the second control signal NRS may be changed from the first level to the second level so that the first control signal NRBS is changed from the second level to the first level before a time point (the time point t**2**) when the transfer control signal TG applied to the transfer transistor TX is changed from the first level to the second level, and the second control signal NRS may be changed from the second level to the first level so that the first control signal NRBS is changed from the first level to the second level after the time point when the transfer control signal TG applied to the transfer transistor TX is changed from the second level to the first level.
[0150] Referring to some example embodiments of FIG. **15**, the first control signal NRBS may maintain the first level during partial sections of the exposure time IT and non-integration time NIT, and the second control signal NRS may maintain the second level during partial sections of the exposure time IT and non-integration time NIT. According to some example embodiments, during the exposure time IT and non-integration time NIT corresponding to sections previous to and following the read time RO of the photodiode, the second transistor NR may be turned on during a partial section and the first transistor NRB may be turned off. Accordingly, noise that may occur during a section aside from the reset time RST and read time RO may be removed or reduced intermittently.
[0151] In the description of the timing diagrams of FIGS. **13** to **15**, a plurality of control signals SEL, RG, TG, NRS, and NRBS described in the timing diagrams may be generated by the row driver **120** of FIG. **2**, and timing control of the plurality of control signals SEL, RG, TG, NRS, and NRBS described in the timing diagrams may be controlled by the timing controller **150** of FIG. **2**.
[0152] Any or all of the elements described with reference to the figures may communicate with any or all other elements described with reference to figures. For example, any element may engage in one-way and/or two-way and/or broadcast communication with any or all other elements in the figures, to transfer and/or exchange and/or receive information such as but not limited to data and/or commands, in a manner such as in a serial and/or parallel manner, via a bus such as a wireless and/or a wired bus (not illustrated). The information may be in encoded various formats, such as in an analog format and/or in a digital format.

[0153] When the terms "about" or "substantially" are used in this specification in connection with a numerical value, it is intended that the associated numerical value includes a manufacturing or operational tolerance (e.g., ±10%) around the stated numerical value. Moreover, when the words "generally" and "substantially" are used in connection with geometric shapes, it is intended that precision of the geometric shape is not required but that latitude for the shape is within the scope of the disclosure. Further, regardless of whether numerical values or shapes are modified as "about" or "substantially," it will be understood that these values and shapes should be construed as including a manufacturing or operational tolerance (e.g., ±10%) around the stated numerical values or shapes.
[0154] As described herein, any electronic devices and/or portions thereof according to any of the example embodiments may include, may be included in, and/or may be implemented by one or more instances of processing circuitry such as hardware including logic circuits; a hardware/software combination such as a processor executing software; or any combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a graphics processing unit (GPU), an application processor (AP), a digital signal processor (DSP), a microcomputer, a field programmable gate array (FPGA), and programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), a neural network processing unit (NPU), an Electronic Control Unit (ECU), an Image Signal Processor (ISP), and the like. In some example embodiments, the processing circuitry may include a non-transitory computer readable storage device (e.g., a memory), for example a DRAM device, storing a program of instructions, and a processor (e.g., CPU) configured to execute the program of instructions to implement the functionality and/or methods performed by some or all of any devices, systems, modules, units, controllers, circuits, architectures, and/or portions thereof according to any of the example embodiments, and/or any portions thereof.
[0155] Hereinabove, some example embodiments have been described in the drawings and specification. In the present specification, although the example embodiments have been described by using specific terms, the terms are used only for descriptive purposes and are not intended to limit the meanings or scope of the inventive concepts described in the claims. Therefore, it will be understood by one of ordinary skill in the art that other modifications and equivalents may be made therein. Accordingly, the scope of the inventive concepts will be defined by the appended claims.

Claims

1. A hybrid pixel comprising: a semiconductor substrate; a pixel circuit including at least one transfer transistor; a sensing circuit configured to detect movement of an object; and at least one photodiode having one end connected to the pixel circuit and another end connected to the sensing circuit, the semiconductor substrate having a first conductivity type, the semiconductor substrate including a photoelectric conversion area corresponding to the at least one photodiode, the semiconductor substrate including a first area of a second conductivity type and a second area surrounding the first area, the second area physically separated from a second area of an adjacent hybrid pixel by a deep trench isolation area, and the semiconductor substrate including the deep trench isolation area.
2. The hybrid pixel of claim 1, wherein the first conductivity type is a p type and the second conductivity type is an n type.
3. The hybrid pixel of claim 1, wherein the deep trench isolation area is a front deep trench isolation.
4. The hybrid pixel of claim 1, wherein a number of the at least one photodiode and a number of the at least one transfer transistor are a same number.
5. The hybrid pixel of claim 4, wherein a number of the first areas corresponds to the number of the

at least one photodiode and a number of the second areas is less than or equal to the number of the first areas.

6. The hybrid pixel of claim 1, wherein the pixel circuit is connected to a cathode of the at least one photodiode and the sensing circuit is connected to an anode of the at least one photodiode.

7. The hybrid pixel of claim 1, wherein the pixel circuit is connected to an anode of the at least one photodiode and the sensing circuit is connected to a cathode of the at least one photodiode.

8. A hybrid pixel comprising: a semiconductor substrate; a pixel circuit including at least one transfer transistor; a sensing circuit configured to detect movement of an object; and at least one photodiode having one end connected to the pixel circuit and another end connected to the sensing circuit, the semiconductor substrate having a first conductivity type, the semiconductor substrate including a photoelectric conversion area corresponding to the at least one photodiode, the semiconductor substrate including a first area of a second conductivity type and a second area surrounding the first area, the second area physically separated from a second area of an adjacent hybrid pixel by a deep trench isolation area, and a first layer including the pixel circuit and the at least one photodiode is different from a second layer including the sensing circuit.

9. The hybrid pixel of claim 8, wherein the first layer is above the second layer, and the hybrid pixel further comprises a vertical area to electrically connect the first layer and the second layer to each other.

10. The hybrid pixel of claim 9, wherein the vertical area is a through silicon via or a chip-to-chip.

11. The hybrid pixel of claim 9, wherein the vertical area comprises: a plurality of first vertical areas electrically connected to each other, corresponding to a plurality of second areas in the first layer; planar areas connecting the plurality of first vertical areas on a plane; and a second vertical area connecting the plurality of first vertical areas and the planar areas to the sensing circuit.

12. The hybrid pixel of claim 9, wherein the deep trench isolation area is a front deep trench isolation.

13. A hybrid sensor comprising: a pixel array including a plurality of hybrid pixels, the plurality of hybrid pixels comprising a photodiode; a pixel circuit connected to one end of the photodiode; a sensing circuit connected to another end of the photodiode; and a noise removal circuit connected between the sensing circuit and the other end of the photodiode, and each of the plurality of hybrid pixels physically separated from an adjacent hybrid pixel by a deep trench isolation area.

14. The hybrid sensor of claim 13, wherein the noise removal circuit comprises: a first transistor connected between the other end of the photodiode and the sensing circuit; and a second transistor connected between the other end of the photodiode and a ground.

15. The hybrid sensor of claim 14, further comprising: a row driver configured to generate signals applied to the pixel array; and a timing controller configured to control timings of the signals generated by the row driver, wherein the timing controller is configured to control timings of a first control signal and a second control signal, which are applied respectively to the first transistor and the second transistor.

16. The hybrid sensor of claim 15, wherein the timing controller is configured to control the timings of the first control signal and the second control signal so that the first transistor is turned off and the second transistor is turned on at a start point of a read time of the photodiode.

17. The hybrid sensor of claim 15, wherein the timing controller is configured to control the timings of the first control signal and the second control signal so that the first transistor is turned on and the second transistor is turned off at an end point of a read time of the photodiode.

18. The hybrid sensor of claim 15, wherein the timing controller is configured to control the timings of the first control signal and the second control signal so that the first transistor is turned on and the second transistor is turned off in sections previous to and following a read time of the photodiode.

19. The hybrid sensor of claim 18, wherein the timing controller is configured to control the timings of the first control signal and the second control signal so that the sections previous to and following the read time of the photodiode include a portion of a section in which the first transistor is turned off and the sections previous to and following the read time of the photodiode include a portion of a section in which the second transistor is turned on.

20. The hybrid sensor of claim 15, wherein the timing controller is configured to control the timings of the first control signal and the second control signal so that the first transistor is turned off and the second transistor is turned on during a reset time of the photodiode.

21.-**22**. (canceled)